1. e3497940
   a) sets of cells having the same color and being on the right/left of the gray column
   b) example of horizontal analogical mappings $\mathcal{I}_k \longrightarrow \mathcal{S}_k$ (between reflected sets). These mappings are also obtain using the action "reflection"
   c) the preserved interactions of the cognitive mappings corresponding to the previous analogical ones are only related to the "vertical position". Therefore, other interactions have to be deleted
2. 3af2c5a8
   a) example with 4 different analogical mappings $\mathcal{I}_k \longrightarrow \mathcal{S}_k$, each one related to a different symmetry (or identity in one case)
   b) If the entities in $\mathcal{I}_k$ are "one set of cells having the same color", then the cognitive mappings defined by the previous analogical mappings do not preserve any interaction (the IS $\mathcal{I}_k$ is actually a singleton set)
   c) If we consider more entities in $\mathcal{I}_k$, then the touching interactions are preserved
3. b60334d2
   a) In $\mathcal{I}_k$ is actually a set of gray cells (there is no interaction)
   b) In $\mathcal{S}_k$ we have "complex squares" as interacting entities, made of cross form + gray vertices + a black interior cell
   c) analogical mappings $\mathcal{I}_k \longrightarrow \mathcal{S}_k$ between gray cells and interior black cells
   d) actions: draw a cross form using blue + complete a cross form into a rectangle using the color of gray cells in $\mathcal{I}_k$
4. 868de0fa
   a) Example of invariance extraction: we have to use vertical analogical/cognitive mapping (there are no interactions) $\mathcal{S}_k \longrightarrow \mathcal{S}_j$ to find that the invariant property is "having even/odd dimensions" (of squares themselves or of interior squares). Note that vertical analogical or cognitive mappings correspond to human's GLOBAL vision of everything in the puzzle.
   b) These invariant properties define conditional action: "if even, draw by red the interior", "if odd, draw by orange the interior"
   c) Note the interaction with the color bar
5. 543a7ed5
   a) Inputs $\mathcal{I}_k$ are sets (no interactions), hence the natural choice for $\mathcal{S}_k$ is also to be a set
   b) The entities on $\mathcal{I}_k$ are complex ones made of rectangles where the sides have a certain thickness (hence interior cells are not necessarily symmetric)
   c) Actions: copy the pink rectangle + draw the minimal rectangle containing the pink one + color interior cells
6. a68b268e
   a) The correct order is violet - light blue - yellow - orange
   b) Entities: set of cells having the same color and placed above/below/right/left the blue row/column
   c) strong (fuzzy) analogical mapping $\mathcal{I}_k \longrightarrow \mathcal{S}_k$ of orange set
   d) weak (fuzzy) analogical mappings $\mathcal{I}_k \longrightarrow \mathcal{S}_k$ of other colored sets
   e) actions: redraw on $\mathcal{S}_k$ the set in $\mathcal{I}_k$ (4 actions)
   f) HiS-ARC constructs an hypothesis for each possible composition of these 4 actions; at each step in this composition (e.g. after only two actions) it evaluates the costs and abandon this hypothesis if the costs are maximal (e.g. orange is not fully preserved); the composition working in all training is the winning hypothesis
7. e509e548
   a) The best interacting entities seem complex entities made of L-forms (sharing some cells)
   b) Using vertical analogical mappings $\mathcal{I}_k \longrightarrow \mathcal{I}_j$, HiS-ARC can discover the invariance property "number of L-forms"
   c) Then, it construct the conditional actions: "If the number of L-forms is one, then redraw in blue", "If the number of L-forms is two, then redraw in violet", "If the number of L-forms is three, then redraw in red"
8. 3906de3d
   a) Entities: set of blue cells + red columns: the latter has to be preferred to other entities because their abstract description `column` is found in every training
   b) By vertical analogical mappings $\mathcal{I}_k \longrightarrow \mathcal{I}_j$, HiS-ARC can also discover the interaction with the black holes/columns in the set of blue cells (and it adds these holes among interacting entities). The interaction is based on the analogies: same horizontal position and same width.
   c) HiS-ARC can also construct strong horizontal analogical mappings $\mathcal{I}_k \longrightarrow S_k$ and hence it can consider actions (moving) taking red columns on $\mathcal{I}_k$ into the analogical ones in $\mathcal{S}_k$.

9. beb8660c
   a) Example of cognitive mappings preserving the relation (i.e. the propagator)
   b) In fact, entities in $\mathcal{I}_k$ can be considered rows having their length in the state
   c) The interactions are $r_1 \xrightarrow{\text{is shorther than}} r_2$ in $\mathcal{I}_k$ and $\bar{r}_1 \xrightarrow{\text{is touching}} \bar{r}_2$
   d) The cognitive maps, constructed starting from the related analogical ones, have to preserve this interactions
   e) HiS-ARC can also consider the right border of the grid and the related touching interaction
10. e179c5f4
   a) $\mathcal{S}_k$ is a linear pattern and hence the analogical/cognitive mappings associate the blue cell in $\mathcal{I}_k$ to the blue patter (made of two diagonals) in $\mathcal{S}_k$
   b) It is also useful to consider the border of the grid and the related touching relations (which are preserved)
   c) We need a general algorithm of the type "complete a linear pattern of arbitrary dimensions x, y and with patter z"
11. a699fb00
   a) Example of linear patterns with black cells
   b) The $\mathcal{I}_k$ and $\mathcal{S}_k$ are isomorphic
12. e98196ab
   a) $\mathcal{I}_k$ is made of two "sets of cells of the same color" separated by a gray row. These sets are not interacting
   b) The mappings are natural and strong and correspond to the output of an action of the type "redraw from $\mathcal{I}_k$ to $\mathcal{S}_k$"
13. 93b581b8
   a) The only entity on the left $\mathcal{I}_k$ is a complex one: "a square made of touching cells of different colors"
   b) The horizontal analogical mapping have only one best possibility (the brown cell on the right corresponds to the brown cell on the left because they are both cells, of the same color and position)
   c) However, in constructing the horizontal cognitive mappings (CEF) $U : \mathcal{I}_k \longrightarrow \mathcal{S}_k$ using the analogical mapping, some entities on $\mathcal{S}_k$ are missed, i.e. the new squares or parts of squares in $\mathcal{S}_k$
   d) Therefore, we have to use RLS starting from each one of these missed entities $\bar{e}$ as patient, i.e. we try to solve the problem: "what are the interactions in $\mathcal{I}_k$ that produce $\bar{e}$ using U?"
   e) Looking (using synaptic plasticity) the transformation that yield squares as output (or parts of them if they go out of the left grid), we can recognize that the light-blue square in $\mathcal{S}_1$, we have the color (from the light blue cell on $\mathcal{I}_1$, but the remaining parameter of this transformation is "position where to start drawing the square".
   f) Using this missing parameter, HiS-ARC is able to compute that we have to add the BLACK cell in (2,2) as interacting entity
   g) HiS-ARC can use RLE to decide the numerical position starting from the position of the light-blue one... but that's cumbersome considering all the other cases
   h) Otherwise, HiS-ARC add new interactions of the form: light-bluebcell, brownbcell, black cell $\xrightarrow{\text{touching on corner}}$ black cell
   i) It adds this interaction because only in this way, the cognitive map U is defined by

$$U(\text{light-blue cell}) = \text{copy of light-blue cell on } \mathcal{S}_1$$

   (by analogical mappings), and

$$U(\text{brown cell}) = \text{copy of brown cell on } \mathcal{S}_1$$
$$U(\text{black cell}, \text{brown cell}, \text{light-blue cell}) = \text{light-blue square in } \mathcal{S}_1$$

   (drawn with the color of the first agent (light-blue cell) and the position of the third agent (black cell)), it gets a CEF
14. 484b58aa
   a) This is a typical example of "completion of a tiling made of (diagonal/vertical/horizontal) patterns"
   b) HiS-ARC firstly recognize "complex diagonals" because they are in every input $\mathcal{I}_k$
   c) After this recognition, it has to search for a higher level entity, such as a "tiling made or diagonal patterns". The idea is to set a general algorithm to find the possible tile pattern (the following ideas seem to work also for tiling made of a pattern repeated in other directions, such as vertical or horizontal).
   d) It can first construct internal analogical mappings $\mathcal{I}_k \longrightarrow \mathcal{I}_k$ by associating analogous diagonals: in this case they have the same colors but different length. For shorter diagonal it will find a low analogy.

e) Then it starts considering a set of cells along a line which is orthogonal to these diagonals by taking one cell in each diagonal until it covers all the diagonals that are *not* analogous: in this way, it considers one cell in each diagonal without taking again cells in already considered diagonals.
f) Then it constructs internal *cognitive* mappings $\mathcal{I}_k \longrightarrow \mathcal{I}_k$ that *preserves* colors and touching relations among cells (or higher entities such as L-forms, rows, small diagonals, etc.?). There is no need to consider all the possible cognitive mappings of this form: we can stop when a fuzzy-based heuristic is sufficiently high or, better, when some cells are missing because the tile pattern is still incomplete.
g) Then it enlarges the set of orthogonal cells (the searching tile pattern) by adding another cell in each diagonal, and restarts constructing internal cognitive mappings.
h) It will stop using a deterministic GEP, i.e. when the minimal set of orthogonal cells is able to reconstruct all the other cells by repeating the discovered pattern (minimal $\{0, 1\}$ value cost or maximal value of exchanged information between the tile pattern and all the other cells, or both).
i) Once it has discovered the tile pattern, the corresponding algorithm "complete the tiling" can start outside the grid and repeat the discovered pattern until the internal analogical mappings between diagonal has a maximal fuzzy value.
j) Using the previous algorithms, HiS-ARC recognize that $\mathcal{S}_k$ is the completion of the incomplete tiling $\mathcal{I}_k$ and hence that $\mathcal{S}_N = \text{completion}(\mathcal{I}_N)$.

15. 6d0160f0
a) Human solution: find the unique square with a yellow cell; redraw on $\mathcal{S}_N$ in the same relative position of the yellow cell.
b) Entities: complex squares + gray rows and columns
c) HiS-ARC can construct an horizontal cognitive mapping in only one way. This mapping correspond to the action "redraw the square X into position (x, y)". It has to decide X and (x, y).
d) Trying to construct vertical analogical mappings, it can find the very weak analogy "both squares have a yellow cell". This could increase its final level of *phasic dopamine signal* (adding different partly positive fuzzy indicators into a final sigmoid function in order to have threshold and saturation effects)
e) On the other hand, it tries to find invariant properties among the squares in $\mathcal{I}_k$ having an analogous one in $\mathcal{S}_k$: the only property it finds is "X has a yellow cell".
f) It also tries to find invariant properties between analogous squares in $\mathcal{I}_k$ and $\mathcal{S}_k$: using the elementary arithmetic module, it can find that the position of the yellow cell is equal to the position of the square in $\mathcal{S}_k$ taking the unit of measurement 3 cells = 1 unit. Therefore, (x, y) = this relative position
g) HiS-ARC constructs the conditional actions: "if X is the square with a yellow cell, then redraw it in the same relative position using the unit 3 cells = 1 unit"

16. 91413438
a) Human solution: repeat the set of colored cells a number of times equals to the number of colored cells. The final grid is 3 times the number of black cells in $\mathcal{I}_k$. For example, $\mathcal{S}_N$ is a square grid of width 3 x 7 = 21, and the light-blue figure has to repeated two times.
b) $\mathcal{I}_k$ is a complex square with touching colored cells and touching black cells or a set of cells having the same color
c) Using the Elementary Arithmetic Unit (EAU), HiS-ARC can discover the dimension of the grid $\mathcal{S}_k$.
d) HiS-ARC can also discover that there are more horizontal cognitive mappings $\mathcal{I}_k \longrightarrow \mathcal{S}_k$. It always count how many of these mappings founds.
e) Using the EAU, it discovers that the number of cognitive mappings equals the number of colored cells in the complex square
f) The action producing these $\mathcal{S}_k$ is hence "Draw the input square $\mathcal{I}_k$ using the linearization order W". Here W can be fixed using hypotheses testing among the few orders that HiS-ARC knows (maybe using synaptic plasticity with all the orders)

17. 7447852a
a) Firstly, HiS-ARC recognizes diagonals, and then a linear pattern. Note that the smallest pattern is made by a black L-form + a red diagonal made of three cells + a black T-form + an opposite diagonal made of two red cells + a single black cell
b) The recognition of the patter is facilitated (increase of dopamine) by the construction of vertical cognitive mappings
c) HiS-ARC sets the pattern as a complex entity made of the previously listed sub-entities. This means that, by translations or using internal cognitive mappings $\mathcal{I}_k \longrightarrow \mathcal{I}_k$, HiS-ARC can enumerate each sub-entites with respect to the order of the pattern: the 1st L-form, the 2nd L-form, the 3rd L-form, etc.

d) HiS-ARC can also complete the founded pattern on an arbitrarily long tiling
e) HiS-ARC constructs horizontal cognitive mappings $\mathcal{I}_k \longrightarrow \mathcal{S}_k$ corresponding to the actions: "color the 1st L-form in yellow", "color the 1+3 th L-form in yellow", "color the 1+3+3 th L-form in yellow"... Similarly for the other sub-entities: "color the 2nd T-form in yellow", "color the 2+3 th T-form in yellow", "color the 2+3+3 T-form in yellow", etc.
f) HiS-ARC can obtain an explanation using also a composition of "translations" and "color in yellow" actions
g) The previous two steps can help the EAU to find the general rule, and hence apply these transformation to $\mathcal{S}_N$

18. 05269061
a) HiS-ARC recognizes a tiling pattern on $\mathcal{S}_k$ and a strong (same object, colors, position) horizontal cognitive mapping
b) Therefore, HiS-ARC recognizes that $\mathcal{S}_k$ is obtained as output of the action: "complete the patter starting from the incomplete X and using the pattern y". Therefore, HiS-ARC needs X and y.
c) HiS-ARC finds the minimal pattern $\bar{y}$ in $\mathcal{S}_k$ and sets $X = \mathcal{S}_k$
d) The entity $\bar{y}$ does not correspond to anything in $\mathcal{I}_k$. HiS-ARC creates a new entity in $\mathcal{I}_k$ using RLS, starting from $\bar{y} \in \mathcal{S}_k$ and finding the best entity in $y \in \mathcal{I}_k$, i.e. one (there are many) having a good fuzzy evaluation (dopamine) in an analogical mapping linking $y \mapsto \bar{y}$. Entities $y \in \mathcal{I}_k$ are three cells diagonal with some black cells.
e) HiS-ARC can now construct vertical analogical mappings $\mathcal{I}_k \longrightarrow \mathcal{I}_j$ that maps pattern $y_k \in \mathcal{I}_k \mapsto y_j \in \mathcal{I}_j$
f) HiS-ARC needs to apply the previous action of completing the pattern on $\mathcal{S}_N$, but it misses the patter $y_N \in \mathcal{I}_N$. Therefore, it uses the previous analogical mappings and set several hypotheses for different colors of the pattern y_N that have a good analogy with y_k. It tests each hypothesis until the "complete the pattern" algorithm works

19. 760b3cac
a) Entities: for the light-blue sets, L-form are not applicable to all training sets (L-forms do not complete these sets and in $\mathcal{I}_3$ there are no L-forms in the light-bleu set). On the contrary, a complex entity made of L-forms (or one L-form and one cell) works for each yellow set. Therefore we have "set of light-blue cells" and "complex entity made of one L-form and ..."
b) There are strong vertical analogical mappings between the complex yellow entities, and two strong horizontal cognitive and analogical mappings. One of the horizontal ones is related to a symmetry action: this action need the x coordinate of a column.
c) Invariance properties extraction has to recognize that the invariant property is the x coordinate of the vertical arm of the yellow L-form + 1. HiS-ARC has now all that is needed to apply the abstract rule to $\mathcal{I}_N$.

20. 0962bcdd
a) Entities in $\mathcal{I}_k$ are complex cross forms. The best entities in $\mathcal{S}_k$ should be "incomplete squares with two cross forms as sub-entities" (but I guess that other entities would work as well)
b) There are strong horizontal cognitive mappings ($\mathcal{I}_k$ is actually isomorphic to $\mathcal{S}_k$)
c) There are also vertical analogical mappings, but probably these are not used
d) HiS-ARC has to discover that entities on $\mathcal{S}_k$ are obtained drawing two cross forms (one of which is complex). Therefore, it needs where (x, y) to start drawing, in what colors c and how much they are long l: these input parameters are taken from $\mathcal{I}_k$ using the EAU

21. eb5a1d5d
a) The simplest entities in $\mathcal{I}_k$ are rectangles with sides of a certain width. Classical squares in $\mathcal{S}_k$. Interactions are "touching rectangles/squares on all interior sides"
b) HiS-ARC can construct horizontal cognitive mappings (they preserve the aforementioned touching interaction)
c) The EAU decides the grid dimension based on the number of colors in input
d) The action that is able to construct $\mathcal{S}_k$ is "draw a square", with inputs "where to draw" and "with what color" taken from input.

22. 995c5fa3
a) $\mathcal{I}_k$ is a (non regular) linear pattern made of squares, U-forms, I-forms and with a black column as separator
b) $\mathcal{S}_k$ is a (non regular) linear pattern made of colored rows
c) Therefore, the most meaningful interactions when dealing with linear patterns are "e_1 precedes e_2 in the linear pattern". The most important property in every linear patter is "position number in the linear pattern", i.e. "the 1st entity, the 2nd entity, the 3rd entity, etc."

d) Weak horizontal analogical mappings based only on the position number. Using these, HiS-ARC discovers that "position number" is an invariant property for the conditional actions "if the position number is n, then draw a row at the same position with color c"
e) Vertical analogical mappings increase the global level of dopamine. The fact that the previous conditional actions works on other inputs also increases the global level of dopamine.
f) This is an example where we can test the idea "add all the (weak) partial fuzzy evaluations (dopamine for single decisions), and insert this sum into a sigmoid function". The output of this function is the global level of dopamine for this HiS-ARC hypothesis. In general, HiS-ARC could have more that one hypotheses and evaluate the global dopamine for all of them in order to decide what is the best hypothesis. If one started hypothesis has a too much low level of global dopamine, then HiS-ARC can stop it and start a new hypothesis in a tree algorithm. The winner of all these hypotheses should have the best evaluation of the GEP (minimal costs and maximal diversification).
g) Note that actually $\mathcal{I}_k \simeq \mathcal{S}_k$.

23. 5117e062
a) Entities of $\mathcal{I}_k$ could be "sets of cells having the same color" and "a single light-blue cell"
b) There is only one horizontal analogical mapping for each training pair
c) These analogy is reinforced by vertical analogical mappings associating light-blue cells at different levels
d) These analogical mapping become cognitive ones if we consider the action: "color the light-blue cell with the color of the other cells touching it" + "copy the other touching cells preserving their color".
e) Therefore, HiS-ARC has to recognize that "a light-blue cell touching other cells of the same color" is an invariant property. This and the previous step allows HiS-ARC to create a conditional action.
f) Note the other entities are forgotten.

24. a8c38be5
a) Entities in $\mathcal{I}_k$ are complex squares having one (zero in one case) L-form or T-form and a set of gray cells as sub-entities.
b) We can construct strong horizontal analogical mappings in one way only
c) It could be useful (for an easiest way of using the EAU) to use a unit of measurement on $\mathcal{S}_k$: 3 cells = 1 unit
d) Of course we have 4 types of L-forms and or T-forms, and each square with the same type is copied (action) in the same position of $\mathcal{S}_k$. Therefore, HiS-ARC has to discover that the invariant property is the type of L-form or T-form and its EAU has to discover this position
e) Using these data, HiS-ARC constructs conditional actions to apply at $\mathcal{I}_N$
f) Note that here we only have two training examples, but even one would suffice (also because included in the final solution, we have to construct the same aforementioned cognitive mappings also for $\mathcal{I}_N$ and $\mathcal{S}_N$: this further increases the global level of dopamine)

25. 855e0971
a) The highest level of entities that HiS-ARC can recognize are "non-regular linear patterns of incomplete rectangles" in $\mathcal{I}_k$, and "non-regular (vertical) linear patterns of (horizontal) linear patterns whose separator is a black (column)"
b) Note that an incomplete rectangle can be recognized by a human whenever we have 4 cells working as vertices. This weak fuzzy evaluation (once again, global dopamine indicator starting from local dopamine indicators) increases if there are L-forms instead of vertices, sides, interior, etc. In this example, humans clearly see "piled rectangles with black holes".
c) Also note that the aforementioned entities in $\mathcal{S}_k$ are recursively constructed: e.g. in $\mathcal{S}_1$, HiS-ARC recognizes that we have two horizontal linear patterns and a rectangle having the same width, and these are the sub-entities of a vertical linear pattern.
d) Using this entities, HiS-ARC cognitive mappings by associating an incomplete rectangle in $\mathcal{I}_k$ with the corresponding linear pattern in $\mathcal{S}_k$: they have the same width and height and this mapping preserves the order in the higher linear pattern: "incomplete light-blue rect. < incomplete yellow rect." is mapped into "light-blue linear pattern < yellow linear pattern".
e) Moreover, the conditional action that corresponds to these mappings is "If the linear patter in $\mathcal{I}_k$ is vertical, then draw vertical column having the same abscissa of the black cell". This abscissa is an invariant property.

26. 85c4e7cd
a) The simplest entities are rectangles
b) The possible interactions are: "completely touching rect." or "r_1 is obtain by r_2 by a scaling of s". The latter defines an order and the EAU has to be able to reverse this order
c) The horizontal cognitive mappings correspond to a coloring action only considering this reversed order.

27. 776ffc46
    a) Human solution: only one figure is inside a square/rectangle (4 sides, different from L-form touching grid's borders!). Find figures of the exact same form among the blue ones. Color them in the color of the internal one.
    b) Simply note that only squares s have an interaction of the form "e is internal to s"
    c) It is also important to construct *internal* analogical mappings of the type $\mathcal{I}_k \longrightarrow \mathcal{I}_k$
28. 3eda0437
    a) Even if HiS-ARC starts only with "sets of contiguous cells of the same color", it can recognize that the violet rectangle in $\mathcal{S}_k$ does not have a counter image along the strong horizontal cognitive mapping.
    b) Therefore, using RLS and analogical mappings $\mathcal{S}_k \longrightarrow \mathcal{I}_k$, it can recognize that a black rectangle entity on $\mathcal{I}_k$ is missing. So, it can add this black rectangle in $\mathcal{I}_k$.
    c) The key point here is the abstract characterization: "the black rectangle in input having the highest area", but that's quite easy using EAU order between numbers. These numbers surely have to include area of rectangles.
29. 6f8cd79b
    a) Let's imagine that in its core-knowledge HiS-ARC only has "rows", "columns" but not "rectangles"
    b) Even with this (paradoxically) poor core-knowledge HiS-ARC recognizes "light-blue touching rows and columns" in $\mathcal{S}_k$.
    c) The, using RLS and horizontal analogical or cognitive mappings, HiS-ARC would start considering "touching black rows and columns" on $\mathcal{I}_k$
    d) Even with this poor core-knowledge, we can imagine that HiS-ARC is able to solve the puzzle
30. 40853293
    a) importance of execution order of action after invariance extraction: to each order corresponds an hypothesis that HiS-ARC tests arriving at a global value of dopamine
31. a699fb00
32. cce03e0d
    a) Entities in $\mathcal{I}_k$ can simply be "sets of (weakly) contiguous cells having the same color". Interactions are "touching"
    b) The key characteristic of this example is the possibility to construct several cognitive mappings, and this can also be useful for HiS-ARC to decide what entities to consider in $\mathcal{S}_k$. For example, if HiS-ARC starts considering "set of weakly contiguous cells having the same color" also in $\mathcal{S}_k$, then it is not able to construct an analogical or cognitive mapping valued in $\mathcal{S}_3$: what is the counter-image of the red (blue) set of cells? On the contrary, HiS-ARC is able to construct several strong cognitive maps starting from the hypothesis of considering in $\mathcal{S}_k$ "sets of cells having the same color". HiS-ARC can also consider "complex squares with interior set of cells having the same color", and this would allow to set a natural unit on $\mathcal{S}_k$. Therefore, some of these hypotheses have a higher level of dopamine.
    c) This level is even more increased by the internal cognitive mappings HiS-ARC is able to construct $\mathcal{S}_k \longrightarrow \mathcal{S}_k$ for $k = 2, 3$.
    d) The horizontal cognitive mappings correspond to the action "copy all the entities of $\mathcal{I}_k$ into $\mathcal{S}_k$ starting from position $(\bar{x}, \bar{y})$"; one action for each cognitive mappings.
    e) The EAU has to discover that: 1) the number of cognitive mappings (and hence of actions) equals the number of red cells. The invariant property are the coordinate (x, y) of each red cell and $\bar{x} = 3x - 2$, $\bar{y} = 3y - 2$.
    f) Another possibility is to try (another HiS-ARC hypothesis?) 3 cells = 1 unit in $\mathcal{S}_k$.
    g) In principle, the presence of copies of the same object in $\mathcal{S}_k$ (or, more generally, of strong internal cognitive mappings $\mathcal{S}_k \longrightarrow \mathcal{S}_k$ for $k = 2, 3$ between objects having the same dimensions (w, h)) is an indicator (synaptic plasticity!) that "in this situation, HiS-ARC can try to use a different unit of measurement.
33. 3e980e27
    a) More than one cognitive mapping: one corresponding to a "copy" action (green cells), and one corresponding to "symmetric" action (red cells). These are therefore conditional actions.
    b) "red cell" and "green cell" are the invariant properties for the conditional actions
34. 4258a5f9
35. ce22a75a
    a) Another example where both analogical mappings of the type $\mathcal{I}_k \longrightarrow \mathcal{S}_k$ and $\mathcal{S}_k \longrightarrow \mathcal{S}_j$ are used to recognize that the best interacting entities in $\mathcal{S}_k$ are "3x3 blue squares (HiS-ARC uses the analogical mappings $\mathcal{S}_1 \longrightarrow \mathcal{S}_2$) centered at the same position of grey cells in $\mathcal{I}_k$. Only after this choice of entities in $\mathcal{S}_2$, HiS-ARC is able to construct the analogical mappings $\mathcal{I}_2 \longrightarrow \mathcal{S}_2$.

36. 321b1fc6
37. 4522001f
    a) This is a good example of hypotheses testing and their evaluations.
    b) The first problem HiS-ARC has is to decide what are entities in the training pairs. In $\mathcal{I}_k$, it has several possibilities, and each one of them represents an hypothesis that need to be tested. The basic idea of the test is "the best hypothesis is one that allows to construct analogical and cognitive mappings, to set invariant properties, to prove RLE or RLS, etc. with the highest value of global dopamine". If, even before the end, i.e. dynamically, HiS-ARC evaluates that the global dopamine of certain set of hypotheses is under a certain threshold, then it starts another set of hypotheses. If all the possible sets of hypotheses (they are always a very limited number) fall under this threshold, then it restarts with the first abandoned one hoping in a future increase of global dopamine.
    c) For example, HiS-ARC has the following possibilities for entities in $\mathcal{I}_k$: L-forms, single cells, grid's vertices, grid's sides, black L-forms
    d) Entities in $\mathcal{S}_k$ are clear: green squares and some entities related to grid, such as grid's vertices.
    e) It seems that the horizontal cognitive mappings can be constructed "only" considering cells in $\mathcal{I}_k$ and adding "black squares" in $\mathcal{S}_k$. Note that these mappings preserves the touching interactions with grid's vertices and sides, but also the touching interactions between cells and squares.
    f) Since all entities in $\mathcal{S}_k$ have the same width and height, HiS-ARC can adopt the 4 cells = 1 unit on $\mathcal{S}_k$ and one grid's side as origin. In this way, the cognitive mappings have a higher level of dopamine (cells' coordinates are preserved into squares' coordinates). Note that, with these units, $\mathcal{S}_k$ grid is 2.25 units of side.
    g) At the end, each cognitive mapping correspond to a clear action, possibly using the EAU.
38. a64e4611
    a) The highest level entities HiS-ARC can construct are "connected sets of cells having the same color". In particular, for the black one, it can also identify it as a complex entity made of rectangles or rows and columns (of a certain width). In this way, HiS-ARC inherits properties and sub-entities such as "interior cells", "border cells", "vertices", "sides" etc.
    b) In this way, the horizontal cognitive mapping is clear and corresponds to the action "color the interior of X using the color c from the color bar"
39. 88a62173
    a) Only one entities allows to construct an horizontal cognitive mapping. Moreover, this entity does not have an internal cognitive mapping $\mathcal{I}_k \longrightarrow \mathcal{I}_k$
    b) In $\mathcal{I}_N$, HiS-ARC tries to construct internal cognitive mappings and, with the only remaining entity, it construct an horizontal cognitive mapping
40. e8593010
    a) Human solution: color entities made of 1 cell in green, those made of two cells in red, those made of three cells in blue
    b) It is important to include "the number of sub-entities" or "the number of cells" as a state property of every entities or, at least, in one of the basic features of the EAU, i.e. "count everything".
    c) Then, HiS-ARC has to identify the "number of cells of X" as the invariant property, and to construct the condition actions we listed in the human solution.
41. f8c80d96
    a) HiS-ARC can proceed in several ways depending on the amplitude of its core-knowledge:
    b) It can identify L-forms and U-forms on $\mathcal{S}_k$, on $\mathcal{I}_k$; then it can identify black entities on the input because corresponding entities on the output do not have a counter-image along the natural cognitive mapping; in this way, it recognize that the action is "color (possibly outside the grid) in gray or light-blue the black forms in input"
    c) It can identify a tiling in output and an horizontal cognitive mapping; therefore, the output can be obtained using the action "complete the tiling" (ones again with possible drawing outside the grid)
42. ac0a08a4
    a) Example of cognitive mapping taking cells into squares and of the use of "unit of measurement": note that of the entities in output has the same dimensions, and that this dimension equals the number of colored squares in input (EAU). Therefore, 1 unit on $\mathcal{S}_k$ = the number of colored cells in $\mathcal{I}_k$"
43. 469497ad
    a) Another example of use of unit of measurement: "the "number of colors" in $\mathcal{I}_k$ corresponds to 2 units in $\mathcal{S}_k$". Analogical mappings are stronger in this way (higher local dopamine), first of all with the red diagonals, see below.

b) The red diagonal in $\mathcal{S}_k$ is an entity without counter-image and leads HiS-ARC to use RLS to insert a "black cell" (1 unit in input corresponds to 1 unit in output) which is in analogy in output $\mathcal{S}_k$ with "a complex entity made of a black square with a diagonal".
c) The corresponding action “draw a red diagonal” can also end up completely outside the output grid

44. d4469b4b
    a) Entities:
        1. Probably there remain only "contiguous sets of cells having the same color" in $\mathcal{I}_k$
        2. In $\mathcal{S}_k$: cross form, T-form, L-form
    b) Interactions: touching (but they’re not important)
    c) Vertical analogical mappings and invariance extraction:
        1. For $\mathcal{I}_k \longrightarrow \mathcal{I}_j$: only the color defines a weak analogy and an invariant property only in some inputs
        2. For $\mathcal{S}_k \longrightarrow \mathcal{S}_j$: strong cognitive mapping, but only between some inputs
    d) This is a typical situation where HiS-ARC has to create conditional actions of the type: “if color $= c$, then draw the form X"
45. d4a91cb9
46. 6aa20dc0
    a) Example of internal analogical mappings $\mathcal{I}_k \longrightarrow \mathcal{I}_k$ and horizontal cognitive mappings $\mathcal{I}_k \longrightarrow S_k$ corresponding to a composition of two actions: “apply symmetry S and then scale of factor f”. Note that there are two possible symmetries involved in the training inputs, and the factor could also be $f = 1$.
47. 5582e5ca
    a) This is a test for the EAU because the final instructions for $\mathcal{I}_N$ involve a characterization of the type: “the color of the set of colored cells having the highest number of cells”
48. 97999447
49. f9012d9b
    a) Another composition of two actions: firstly complete the pattern, then draw in output only the part corresponding the the black square
50. d687bc17
    a) HiS-ARC can consider interaction of the type "having the same color" or internal analogical mappings based on the same criteria.
    b) Invariance extraction leads to conditional action of the type: "if in analogy with a row, draw a touching cell having the same x" or "if in analogy with a column, draw a touching cell having the same y"
    c) Cells that are not interacting or having zero analogy are simply forgotten.